SUDARSAN SRINIVASAN

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SUMMARY

- Mechanical Engineer with experience in consumer product development and automotive powertrain design
- Comfortable tackling ambiguous problems from first principles, from root cause investigation through to production
- Skilled in CAD modeling (Siemens NX, SolidWorks), GD&T, tolerance stack-up analysis, and FEA, with hands-on experience across die casting, injection molding, and sheet metal, driving design optimization from concept through high-volume production

EXPERIENCE

SharkNinja ***Jan 2026 – Jun 2026***

Product Design Engineering Co-op, New Product Development *Needham, MA*

- Drove root cause investigation on compressor stall failures across 5 units; identified debris jamming on the compressor rotor through systematic teardowns, coordinated with China supplier for batch-level RCA, and supported an interim software correction
- Owned risk assessment for a design simplification on a critical slush machine component; ran performance trials across key recipes, confirmed design parity on 75% of test cases, and implemented a software adjustment to address remaining gaps
- Built a data analysis tool for processing 73-column raw data logs from frozen treats machines: automated data cleaning, anomaly flagging, and chart generation, cutting analysis time from 15 minutes to under 10 seconds per run
- Developed a rapid condensation test method using a mist-spray setup as an alternative to an unavailable humidity chamber, enabling prototype validation for condensation management while reducing test cycle time by ~80%

The Boring Company ***Sep 2025 - Dec 2025***

Integration Engineering Intern *Bastrop, TX*

- Led subsystem integration for the MB-1 TBM, coordinating across teams to design and integrate ventilation, fluid distribution, and gantry systems; resolved build-blocking issues including part kitting errors, routing conflicts, and supply delays to keep builds on schedule
- Supported integration of a 2,000+ part Vertical Cassette sub-assembly over two months by validating part readiness, resolving engineering discrepancies, and organizing build documentation and technician workflows, helping keep the assembly on schedule

Bajaj Auto Limited ***Jan 2021 - Jul 2024***

Mechanical Design Engineer, Powertrain Design *Pune, India*

- Owned design of aluminum die-cast (HPDC) components, injection-molded covers, sheet-metal brackets, and gaskets for ICE and EV powertrain platforms, applying GD&T and tolerance-stack-up analysis from concept through DFMEA, DVP, and high-volume production launch
- Directed 3D CAD modeling and 2D drawing creation in Siemens NX, reviewing designs for intent, tight tolerance control, and dimensional accuracy across HPDC, machined, and injection-molded components, resulting in reduced design rework and higher fit-check success
- Applied FEA simulations in ANSYS for over 20 design studies, performing static structural and contact-pressure analyses to validate component integrity and sealing performance before physical prototyping, which eliminated the need for multiple prototype iterations
- Led multiple VAVE initiatives across powertrain components, including part rationalization, material substitution, and design optimization, cumulatively reducing component mass and delivering $15K+ in annual cost savings

- Automated gasket CAD modeling and validation workflows using NXOpen scripting, cutting design cycle time by 76%
- Drove root cause analysis of 100+ in-house and field failures using 5 Whys and FTA, collaborated with quality, testing, and supply-chain teams, and implemented corrective actions that prevented recurrence of similar failures

TECHNICAL SKILLS

- **CAD & Simulation:** Siemens NX, SolidWorks, ANSYS Mechanical, FEA
- **Mechanical Design and Analysis**: GD&T, Tolerance Stack-up Analysis, DFM/DFA, DFMEA, Root Cause Analysis, DOE
- **Manufacturing and Prototyping**: Machining, 3D Printing, Die Casting, Injection Molding, Sheet Metal, HPDC, BOM
- **Programming:** MATLAB, Python
- **PLM and Tools:** Siemens Teamcenter

PROJECTS

Boeing Advanced Research Collaboration, University of Washington ***Jan 2025 – Dec 2025***

Autonomous Cleaning Robot, Mechanical Subsystem Integration *Seattle, WA*

- Led the mechanical architecture of an autonomous cleaning robot for an industrial sump environment, owning chassis design, component packaging, and enclosure development within tight height, weight, and wet-environment constraints
- Generated multiple cleaning system concepts and built test fixtures to physically compare cleaning effectiveness on the inclined epoxy surface, down-selecting through a Pugh matrix to the most feasible solution for wet, low-maintenance operation
- Designed the selected cleaning system from concept to detail, defining mechanism geometry, mounting, and component interfaces ahead of procurement and integration
- Developed CAD assemblies to validate fit, alignment, and interface compatibility, iterating on physical prototypes with material and component tradeoffs to balance durability, assembly simplicity, and field usability

EDUCATION

University of Washington, Seattle, WA *Sep 2024 - Aug 2026*

Master of Science, Mechanical Engineering *GPA: 3.90*

National Institute of Technology, Trichy, India *Jun 2016 - Jun 2020*

Bachelor of Technology in Mechanical Engineering (Minor: Electrical and Electronics)

CERTIFICATIONS

Certified SolidWorks Professional (CSWP)